

## Qualification Pack



# Pulses Cultivator

QP Code: AGR/Q0104

Version: 4.0

NSQF Level: 3

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## Qualification Pack

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## Qualification Pack

### AGR/Q0104: Pulses Cultivator

#### Brief Job Description

A Pulses Cultivator undertakes the cultivation of pulses at the farm level. The individual is also responsible for harvesting and marketing the pulses. The person also undertakes integrated pest and disease management, along with weed management as part of the job role.

#### Personal Attributes

The individual must be physically fit to work for long durations. The person must be able to take decisions independently and work in coordination with others. The abilities to read, write and communicate well are the other important attributes required in this job role.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [AGR/N0118: Select and prepare the site and sow the pulse seeds](#)
2. [AGR/N0108: Carry out macro and micronutrient management for field crops](#)
3. [AGR/N0109: Manage weed growth in crop fields](#)
4. [AGR/N0110: Perform integrated pest and disease management for pulses](#)
5. [AGR/N0111: Perform irrigation management for field crops](#)
6. [AGR/N0120: Carry out harvesting, post-harvest processing and marketing of pulses](#)
7. [AGR/N9933: Participate in collective farming activities](#)
8. [AGR/N9932: Follow health and safety guidelines](#)
9. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

#### Qualification Pack (QP) Parameters

|                   |                                      |
|-------------------|--------------------------------------|
| <b>Sector</b>     | Agriculture                          |
| <b>Sub-Sector</b> | Agriculture Crop Production          |
| <b>Occupation</b> | Field Crops Cultivation (Food Crops) |

## Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Country</b>                                            | India                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>NSQF Level</b>                                         | 3                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Credits</b>                                            | 9                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NCO-2015/6111.0401                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Minimum Educational Qualification &amp; Experience</b> | 10th grade pass (or equivalent)<br>OR<br>8th grade pass with 3 Years of experience relevant experience in agriculture and allied sectors<br>OR<br>Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience relevant experience in agriculture and allied sectors<br>OR<br>Previous relevant Qualification of NSQF Level (3) with 3 Years of experience relevant experience in agriculture and allied sectors |
| <b>Minimum Level of Education for Training in School</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Minimum Job Entry Age</b>                              | 18 Years                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Next Review Date</b>                                   | 18/02/2028                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>NSQC Approval Date</b>                                 | 18/02/2025                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Version</b>                                            | 4.0                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Reference code on NQR</b>                              | QG-03-AG-03537-2025-V2-ASCI                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>NQR Version</b>                                        | 2                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## Qualification Pack

### AGR/N0118: Select and prepare the site and sow the pulse seeds

#### Description

This OS unit is about selecting and preparing the site for the cultivation of pulses along with procuring and sowing the pulse seeds. It also covers resource optimisation, waste management and practising inclusion at work.

#### Scope

The scope covers the following :

- Select and prepare the site for the cultivation of pulses
- Procure and prepare the pulse seeds for sowing
- Prepare the field
- Sow the pulse seeds
- Optimise resource utilisation
- Perform waste management

#### Elements and Performance Criteria

##### *Select and prepare the site for the cultivation of pulses*

To be competent, the user/individual on the job must be able to:

- PC1.** check the site has conducive climatic conditions for the cultivation of pulses and is free from limiting factors such as flooding, drought, extreme heat and cold
- PC2.** coordinate with an authorised lab to determine if the soil is suitable for the cultivation of pulses
- PC3.** ensure the site is accessible and has availability of quality water, labour and other inputs
- PC4.** identify the risks associated with the cultivation of pulses at the site and take appropriate preventive measures

##### *Procure and prepare the pulse seeds for sowing*

To be competent, the user/individual on the job must be able to:

- PC5.** select the appropriate variety of pulses to be cultivated based on the climate, soil condition, time of planting, cultivation duration, marketability, etc.
- PC6.** procure high quality certified pulse seeds with resistance to local pests and diseases from a government-approved source in the required quantity
- PC7.** treat the pulse seeds using the recommended biofertilizer, insecticide, pesticide or fungicide in the prescribed dose
- PC8.** remove and dispose the damaged/ diseased seeds
- PC9.** store the treated seeds at the recommended temperature and humidity, ensuring hygienic conditions in the storage area

##### *Prepare the field*

To be competent, the user/individual on the job must be able to:

- PC10.** remove all the weeds and waste materials from the field
- PC11.** identify the soil type and carry out ploughing in the field to achieve the required tilt

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- PC12.** level the field appropriately to ensure uniformly graded field
- PC13.** create ridges and furrows in the field to avoid waterlogging
- PC14.** apply the recommended organic manure in the field in an appropriate quantity
- PC15.** check the soil's pH level using the pH meter and apply lime, gypsum or other relevant treatment in an appropriate quantity to adjust the pH
- PC16.** adopt zero tillage or minimum tillage practices where feasible to conserve soil moisture

### *Sow the pulse seeds*

To be competent, the user/individual on the job must be able to:

- PC17.** follow the recommended sowing period for the chosen pulse crop based on climatic condition
- PC18.** select an appropriate method for sowing pulse seeds such as broadcasting or mechanised sowing
- PC19.** sow the pulse seeds at the appropriate seed rate using the relevant tools and implements
- PC20.** maintain the recommended planting depth and density to achieve the healthy growth of plants
- PC21.** carry out intercropping with a suitable crop to achieve a higher yield
- PC22.** integrate pulses into crop rotations to enhance soil fertility and reduce pest build-up

### *Optimise resource utilisation*

To be competent, the user/individual on the job must be able to:

- PC23.** optimise the usage of water and other resources in various tasks and processes
- PC24.** plug water leakages to prevent its wastage

### *Perform waste management*

To be competent, the user/individual on the job must be able to:

- PC25.** segregate waste into appropriate categories
- PC26.** recycle the recyclable waste appropriately and dispose the non-recyclable waste in an environment-friendly manner

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** various pulse varieties and hybrids such as green gram, red gram, Bengal gram, beans, peas, etc.
- KU2.** the criteria for selecting a site for the cultivation of pulses
- KU3.** various agro-climatic zones in India suitable for the cultivation of pulses
- KU4.** the site, climate, soil type, soil fertility, nature of subsoil and soil depth suited for growing pulses
- KU5.** the process of getting the soil sample tested through an authorised lab to determine the soil's suitability for the cultivation of pulses
- KU6.** various inputs required for the cultivation of pulses such as water, fertilisers, pesticides, labour, etc.
- KU7.** the use of various tools and implements for the preparation of field for the cultivation of pulses, such as chisel plough, disc plough, subsoiler, tiller, harrow, cultivator, etc.
- KU8.** the importance of preparing the field according to the soil type

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- KU9.** the appropriate tilth required for sowing pulse seeds
- KU10.** the importance of levelling the field for uniform and efficient irrigation
- KU11.** the benefits of creating ridges and furrows
- KU12.** how to use a pH meter and adjust the soil's pH
- KU13.** the criteria for selecting pulses varieties to cultivate such as yield quantity and period, climate along with resistance to various pests, diseases and abiotic stress, etc.
- KU14.** the appropriate time for sowing pulse seeds based on the moisture content in the soil, precipitation, humidity, etc.
- KU15.** the importance of procuring seeds from a government-approved source and the prevalent market prices for different varieties
- KU16.** the process of treating pulse seeds before sowing
- KU17.** the appropriate temperature and humidity for storing the pulse seeds before and after treatment
- KU18.** how to recognise hard and diseased pulse seeds
- KU19.** the recommended seed rate for different varieties of pulses
- KU20.** different methods for sowing pulse seeds such as broadcasting and mechanised sowing
- KU21.** use of the relevant tools and implements for sowing pulse seeds
- KU22.** recommended planting density to be maintained while sowing pulse seeds
- KU23.** different crops suitable for intercropping with pulses
- KU24.** benefits of resource optimisation and different ways to optimise resources
- KU25.** the criteria for segregating waste into appropriate categories
- KU26.** how to recycle and dispose different types of waste

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write work-related notes and maintain the relevant records
- GS2.** read the relevant literature to get the latest updates about the field of work
- GS3.** listen attentively to understand the information/ instructions being shared
- GS4.** communicate politely and professionally
- GS5.** plan and prioritise tasks to ensure timely completion
- GS6.** evaluate all possible solutions to a problem to select the best one
- GS7.** co-ordinate with the co-workers to achieve the work objectives
- GS8.** identify possible disruptions to work and take appropriate preventive measures
- GS9.** take quick decisions to deal with workplace emergencies/ accidents

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Select and prepare the site for the cultivation of pulses</i>                                                                                                              | <b>4</b>     | <b>6</b>        | -             | <b>4</b>   |
| <b>PC1.</b> check the site has conducive climatic conditions for the cultivation of pulses and is free from limiting factors such as flooding, drought, extreme heat and cold | -            | -               | -             | -          |
| <b>PC2.</b> coordinate with an authorised lab to determine if the soil is suitable for the cultivation of pulses                                                              | -            | -               | -             | -          |
| <b>PC3.</b> ensure the site is accessible and has availability of quality water, labour and other inputs                                                                      | -            | -               | -             | -          |
| <b>PC4.</b> identify the risks associated with the cultivation of pulses at the site and take appropriate preventive measures                                                 | -            | -               | -             | -          |
| <i>Procure and prepare the pulse seeds for sowing</i>                                                                                                                         | <b>4</b>     | <b>4</b>        | -             | <b>4</b>   |
| <b>PC5.</b> select the appropriate variety of pulses to be cultivated based on the climate, soil condition, time of planting, cultivation duration, marketability, etc.       | -            | -               | -             | -          |
| <b>PC6.</b> procure high quality certified pulse seeds with resistance to local pests and diseases from a government-approved source in the required quantity                 | -            | -               | -             | -          |
| <b>PC7.</b> treat the pulse seeds using the recommended biofertilizer, insecticide, pesticide or fungicide in the prescribed dose                                             | -            | -               | -             | -          |
| <b>PC8.</b> remove and dispose the damaged/ diseased seeds                                                                                                                    | -            | -               | -             | -          |
| <b>PC9.</b> store the treated seeds at the recommended temperature and humidity, ensuring hygienic conditions in the storage area                                             | -            | -               | -             | -          |
| <i>Prepare the field</i>                                                                                                                                                      | <b>6</b>     | <b>10</b>       | -             | <b>8</b>   |
| <b>PC10.</b> remove all the weeds and waste materials from the field                                                                                                          | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                         | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> identify the soil type and carry out ploughing in the field to achieve the required tilt                                                    | -            | -               | -             | -          |
| <b>PC12.</b> level the field appropriately to ensure uniformly graded field                                                                              | -            | -               | -             | -          |
| <b>PC13.</b> create ridges and furrows in the field to avoid waterlogging                                                                                | -            | -               | -             | -          |
| <b>PC14.</b> apply the recommended organic manure in the field in an appropriate quantity                                                                | -            | -               | -             | -          |
| <b>PC15.</b> check the soil's pH level using the pH meter and apply lime, gypsum or other relevant treatment in an appropriate quantity to adjust the pH | -            | -               | -             | -          |
| <b>PC16.</b> adopt zero tillage or minimum tillage practices where feasible to conserve soil moisture                                                    | -            | -               | -             | -          |
| <i>Sow the pulse seeds</i>                                                                                                                               | <b>6</b>     | <b>4</b>        | -             | <b>6</b>   |
| <b>PC17.</b> follow the recommended sowing period for the chosen pulse crop based on climatic condition                                                  | -            | -               | -             | -          |
| <b>PC18.</b> select an appropriate method for sowing pulse seeds such as broadcasting or mechanised sowing                                               | -            | -               | -             | -          |
| <b>PC19.</b> sow the pulse seeds at the appropriate seed rate using the relevant tools and implements                                                    | -            | -               | -             | -          |
| <b>PC20.</b> maintain the recommended planting depth and density to achieve the healthy growth of plants                                                 | -            | -               | -             | -          |
| <b>PC21.</b> carry out intercropping with a suitable crop to achieve a higher yield                                                                      | -            | -               | -             | -          |
| <b>PC22.</b> integrate pulses into crop rotations to enhance soil fertility and reduce pest build-up                                                     | -            | -               | -             | -          |
| <i>Optimise resource utilisation</i>                                                                                                                     | <b>4</b>     | <b>8</b>        | -             | <b>4</b>   |
| <b>PC23.</b> optimise the usage of water and other resources in various tasks and processes                                                              | -            | -               | -             | -          |
| <b>PC24.</b> plug water leakages to prevent its wastage                                                                                                  | -            | -               | -             | -          |

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| Assessment Criteria for Outcomes                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Perform waste management</i>                                                                                                | 6            | 8               | -             | 4          |
| <b>PC25.</b> segregate waste into appropriate categories                                                                       | -            | -               | -             | -          |
| <b>PC26.</b> recycle the recyclable waste appropriately and dispose the non-recyclable waste in an environment-friendly manner | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                               | <b>30</b>    | <b>40</b>       | <b>-</b>      | <b>30</b>  |

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### National Occupational Standards (NOS) Parameters

|                            |                                                     |
|----------------------------|-----------------------------------------------------|
| <b>NOS Code</b>            | AGR/N0118                                           |
| <b>NOS Name</b>            | Select and prepare the site and sow the pulse seeds |
| <b>Sector</b>              | Agriculture                                         |
| <b>Sub-Sector</b>          | Agriculture Crop Production                         |
| <b>Occupation</b>          | Field Crops Cultivation (Food Crops)                |
| <b>NSQF Level</b>          | 3                                                   |
| <b>Credits</b>             | 1                                                   |
| <b>Version</b>             | 4.0                                                 |
| <b>Last Reviewed Date</b>  | 18/02/2025                                          |
| <b>Next Review Date</b>    | 18/02/2028                                          |
| <b>NSQC Clearance Date</b> | 18/02/2025                                          |

## Qualification Pack

### AGR/N0108: Carry out macro and micronutrient management for field crops

#### Description

This OS unit is about managing the macro and micronutrient needs of field crops to ensure their optimum growth

#### Scope

The scope covers the following :

- Determine the macro and micronutrients requirements
- Apply fertilisers to the soil
- Perform soil conservation

#### Elements and Performance Criteria

##### *Determine the macro and micronutrients requirements*

To be competent, the user/individual on the job must be able to:

- PC1.** coordinate with an authorised lab to get the soil sample tested, ensuring the sample is collected from the field, packed and labelled as per the applicable procedure
- PC2.** interpret the soil analysis report to determine the micro and macronutrients requirements of the soil based on the planned crop variety or coordinate with an expert for the purpose
- PC3.** select the appropriate organic and inorganic fertilisers, ensuring they contain the required nutrients in the recommended quantity
- PC4.** prepare organic fertilisers such as farmyard manure, vermicompost and inorganic fertiliser solutions, ensuring personal safety

##### *Apply fertilisers to the soil*

To be competent, the user/individual on the job must be able to:

- PC5.** prepare the mixture of liquid fertilisers for application in the field, using them in the recommended quantity
- PC6.** prepare the field for the application of fertilisers
- PC7.** apply organic and inorganic fertilisers containing the required macro and micronutrients to the soil in the recommended dose
- PC8.** use nitrogen-fixing bacteria (e.g., Rhizobium inoculants) to improve nitrogen availability in the soil
- PC9.** regulate the dose of fertiliser according to the crop cycle
- PC10.** maintain the record of fertilisers used in the field

##### *Perform soil conservation*

To be competent, the user/individual on the job must be able to:

- PC11.** prepare a soil nutrition supplementation calendar based on the stages of the crop's growth
- PC12.** follow the recommended practices for soil conservation, such as mulching, crop rotation or planting cover crops to preserve soil health, prevent erosion and application of organic fertilizers

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**PC13.** follow the regenerative practices like no-till farming, cover cropping, and use of biofertilizers

### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the basic concepts of plant nutrition and soil fertility
- KU2.** different types of macro and micronutrients, their properties and functions
- KU3.** common symptoms of nutrient deficiency in plants
- KU4.** the use of different types of green manure and nitrogen-fixing crops
- KU5.** the process of soil sampling and testing
- KU6.** how to interpret the soil analysis report to determine the macro and micronutrient requirements of the soil
- KU7.** different soil types, their advantages and disadvantages with reference to the presence of various nutrients
- KU8.** how to prepare the mixture of liquid fertilisers, using them in the recommended quantity
- KU9.** how to prepare the field for the application of fertilisers
- KU10.** the appropriate time, methods and dose for the application of different types of fertilisers for a variety of crops
- KU11.** the importance of regulating the dose of fertiliser according to the crop cycle
- KU12.** importance of soil conservation and various soil conservation practices
- KU13.** the importance of getting the soil tested through a government-approved soil-testing laboratory to determine the macro and micronutrients present in it, and the requirement of adding the nutrients manually
- KU14.** varieties of organic and inorganic fertilisers to be applied to the soil to improve its fertility, and the nutrient content in them
- KU15.** the process of preparing organic fertilisers such as farmyard manure, vermicompost and inorganic fertiliser solutions
- KU16.** how to apply organic and inorganic fertilisers to the soil and the recommended dose to be used
- KU17.** the harmful effects of over-dosage of fertilisers
- KU18.** applicable documentation requirements
- KU19.** the process of preparing a soil nutrition supplementation calendar based on the stages of the crop's growth

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** communicate politely and professionally
- GS3.** read the relevant literature to learn about the latest developments in the field of work
- GS4.** listen attentively to understand the information/ instructions being shared

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- GS5.** plan and prioritise tasks to ensure timely completion
- GS6.** co-ordinate with the co-workers to achieve the work objectives
- GS7.** evaluate all possible solutions to a problem to select the best one
- GS8.** identify possible disruptions to work and take appropriate preventive measures
- GS9.** take quick decisions to deal with workplace emergencies/ accidents

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Determine the macro and micronutrients requirements</i>                                                                                                                                       | <b>12</b>    | <b>10</b>       | -             | <b>8</b>   |
| <b>PC1.</b> coordinate with an authorised lab to get the soil sample tested, ensuring the sample is collected from the field, packed and labelled as per the applicable procedure                | -            | -               | -             | -          |
| <b>PC2.</b> interpret the soil analysis report to determine the micro and macronutrients requirements of the soil based on the planned crop variety or coordinate with an expert for the purpose | -            | -               | -             | -          |
| <b>PC3.</b> select the appropriate organic and inorganic fertilisers, ensuring they contain the required nutrients in the recommended quantity                                                   | -            | -               | -             | -          |
| <b>PC4.</b> prepare organic fertilisers such as farmyard manure, vermicompost and inorganic fertiliser solutions, ensuring personal safety                                                       | -            | -               | -             | -          |
| <i>Apply fertilisers to the soil</i>                                                                                                                                                             | <b>10</b>    | <b>16</b>       | -             | <b>10</b>  |
| <b>PC5.</b> prepare the mixture of liquid fertilisers for application in the field, using them in the recommended quantity                                                                       | -            | -               | -             | -          |
| <b>PC6.</b> prepare the field for the application of fertilisers                                                                                                                                 | -            | -               | -             | -          |
| <b>PC7.</b> apply organic and inorganic fertilisers containing the required macro and micronutrients to the soil in the recommended dose                                                         | -            | -               | -             | -          |
| <b>PC8.</b> use nitrogen-fixing bacteria (e.g., Rhizobium inoculants) to improve nitrogen availability in the soil                                                                               | -            | -               | -             | -          |
| <b>PC9.</b> regulate the dose of fertiliser according to the crop cycle                                                                                                                          | -            | -               | -             | -          |
| <b>PC10.</b> maintain the record of fertilisers used in the field                                                                                                                                | -            | -               | -             | -          |
| <i>Perform soil conservation</i>                                                                                                                                                                 | <b>8</b>     | <b>14</b>       | -             | <b>12</b>  |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> prepare a soil nutrition supplementation calendar based on the stages of the crop's growth                                                                                                      | -            | -               | -             | -          |
| <b>PC12.</b> follow the recommended practices for soil conservation, such as mulching, crop rotation or planting cover crops to preserve soil health, prevent erosion and application of organic fertilizers | -            | -               | -             | -          |
| <b>PC13.</b> follow the regenerative practices like no-till farming, cover cropping, and use of biofertilizers                                                                                               | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                             | <b>30</b>    | <b>40</b>       | <b>-</b>      | <b>30</b>  |

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### National Occupational Standards (NOS) Parameters

|                            |                                                              |
|----------------------------|--------------------------------------------------------------|
| <b>NOS Code</b>            | AGR/N0108                                                    |
| <b>NOS Name</b>            | Carry out macro and micronutrient management for field crops |
| <b>Sector</b>              | Agriculture                                                  |
| <b>Sub-Sector</b>          | Agriculture Crop Production                                  |
| <b>Occupation</b>          | Field Crops Cultivation (Food Crops)                         |
| <b>NSQF Level</b>          | 4                                                            |
| <b>Credits</b>             | 1                                                            |
| <b>Version</b>             | 4.0                                                          |
| <b>Last Reviewed Date</b>  | 08/05/2025                                                   |
| <b>Next Review Date</b>    | 31/10/2025                                                   |
| <b>NSQC Clearance Date</b> | 08/05/2025                                                   |

## Qualification Pack

### AGR/N0109: Manage weed growth in crop fields

#### Description

This OS unit is about managing the growth of weeds in crop fields. It covers both preventive and remedial measures for weed control.

#### Scope

The scope covers the following :

- Identify weed growth
- Perform weed management

#### Elements and Performance Criteria

##### *Identify weed growth*

To be competent, the user/individual on the job must be able to:

- PC1.** inspect the field, bunds, thrashing yard, granary, composting area, and irrigation and drainage channels periodically to identify weed growth
- PC2.** maintain the record of observations with respect to weed identification and their growth

##### *Perform weed management*

To be competent, the user/individual on the job must be able to:

- PC3.** select an appropriate combination of different types of intercultural and mechanical methods for effective weed control such as trash mulching, solarisation and pasteurisation
- PC4.** select and prepare the recommended herbicide/ bio-herbicide solution suitable to the crop
- PC5.** spray the herbicide/ bio-herbicide safely in the recommended dose
- PC6.** remove weeds manually using the appropriate hand tools and implements, as required
- PC7.** retain the weeds of importance during the weeding process
- PC8.** maintain the herbicides and herbicide application equipment separately to prevent cross-contamination with other chemicals
- PC9.** use organic mulches to suppress weed growth and conserve soil moisture

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the critical period for organic weed control, reducing the dependence on herbicides and weedicides
- KU2.** the adverse effect of different types of weed on crop growth such as grass, broad leaves, sedges
- KU3.** different weed control methods such as preventative, intercultural, mechanical, biological and chemicals
- KU4.** advantages and disadvantages of different weeding methods
- KU5.** the critical period of crop-weed competition

## Qualification Pack

- KU6.** different manual weeding techniques
- KU7.** the use of relevant weeding equipment such as hoe and spade
- KU8.** use of pre-emergent and post-emergent herbicides
- KU9.** differences between blanket and spot application of herbicides
- KU10.** the process of soil solarisation and pasteurisation
- KU11.** environmental norms to be adhered to during herbicide application
- KU12.** the effect of herbicide residue on different types of crop
- KU13.** ways to minimise pollution caused by overuse of herbicides
- KU14.** the importance of inspecting the field regularly to identify weed growth
- KU15.** the appropriate combination of different types of intercultural and mechanical methods for effective weed control such as solarisation and pasteurisation
- KU16.** the process of selecting and preparing the recommended herbicide/ bio-herbicide solution suitable to the crop
- KU17.** how to spray herbicide/ bio-herbicide safely on the crop
- KU18.** the importance of retaining the weeds of importance during the weeding process
- KU19.** the importance of maintaining the herbicides and herbicide application equipment separately to prevent cross-contamination with other chemicals

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** read the relevant guides, manuals and literature to get the latest updates about the field of work
- GS3.** communicate clearly and politely
- GS4.** listen attentively to understand the instructions being given
- GS5.** identify effective solutions to work-related issues
- GS6.** plan and prioritise tasks to ensure timely completion
- GS7.** take quick decisions in case of an emergency/ accident
- GS8.** plan effective use of time and resources

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                            | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Identify weed growth</i>                                                                                                                                                                 | <b>10</b>    | <b>12</b>       | -             | <b>14</b>  |
| <b>PC1.</b> inspect the field, bunds, thrashing yard, granary, composting area, and irrigation and drainage channels periodically to identify weed growth                                   | -            | -               | -             | -          |
| <b>PC2.</b> maintain the record of observations with respect to weed identification and their growth                                                                                        | -            | -               | -             | -          |
| <i>Perform weed management</i>                                                                                                                                                              | <b>20</b>    | <b>28</b>       | -             | <b>16</b>  |
| <b>PC3.</b> select an appropriate combination of different types of intercultural and mechanical methods for effective weed control such as trash mulching, solarisation and pasteurisation | -            | -               | -             | -          |
| <b>PC4.</b> select and prepare the recommended herbicide/ bio-herbicide solution suitable to the crop                                                                                       | -            | -               | -             | -          |
| <b>PC5.</b> spray the herbicide/ bio-herbicide safely in the recommended dose                                                                                                               | -            | -               | -             | -          |
| <b>PC6.</b> remove weeds manually using the appropriate hand tools and implements, as required                                                                                              | -            | -               | -             | -          |
| <b>PC7.</b> retain the weeds of importance during the weeding process                                                                                                                       | -            | -               | -             | -          |
| <b>PC8.</b> maintain the herbicides and herbicide application equipment separately to prevent cross- contamination with other chemicals                                                     | -            | -               | -             | -          |
| <b>PC9.</b> use organic mulches to suppress weed growth and conserve soil moisture                                                                                                          | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                            | <b>30</b>    | <b>40</b>       | -             | <b>30</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                      |
|----------------------------|--------------------------------------|
| <b>NOS Code</b>            | AGR/N0109                            |
| <b>NOS Name</b>            | Manage weed growth in crop fields    |
| <b>Sector</b>              | Agriculture                          |
| <b>Sub-Sector</b>          | Agriculture Crop Production          |
| <b>Occupation</b>          | Field Crops Cultivation (Food Crops) |
| <b>NSQF Level</b>          | 4                                    |
| <b>Credits</b>             | 1                                    |
| <b>Version</b>             | 4.0                                  |
| <b>Last Reviewed Date</b>  | 08/05/2025                           |
| <b>Next Review Date</b>    | 31/10/2025                           |
| <b>NSQC Clearance Date</b> | 08/05/2025                           |

## Qualification Pack

### AGR/N0110: Perform integrated pest and disease management for pulses

#### Description

This OS unit is about performing integrated pest management for pulses and taking various measures to prevent and control insects, pests, and diseases in it.

#### Scope

The scope covers the following :

- Follow the relevant preventive measures
- Identify pests and disease in pulses
- Identify and apply the necessary treatment

#### Elements and Performance Criteria

##### *Follow the relevant preventive measures*

To be competent, the user/individual on the job must be able to:

- PC1.** use pest and disease-resistant varieties of pulses
- PC2.** carry out crop rotation with suitable crops
- PC3.** follow the recommended practices to restrict the entry of pathogens into the field through planting material, irrigation water, workers, tools and equipment, and vectors such as whitefly
- PC4.** follow the relevant preventive techniques according to the cultivar such as light trap, bird perch, sticky traps, etc.
- PC5.** identify and remove the diseased crop to prevent the spread of pest and disease to healthy crop
- PC6.** use the recommended combination of biological, mechanical and chemical control methods for effective pest and disease prevention

##### *Identify pests and disease in pulses*

To be competent, the user/individual on the job must be able to:

- PC7.** follow the recommended Integrated Pest Management (IPM) methods such as light and pheromone traps to identify the presence and population of pests, insects and vectors such as pod borers, hairy caterpillar, aphids, stem fly, whitefly, thrips, beetles, nematodes, etc.
- PC8.** conduct regular field visits to identify major pulse diseases such as leaf spot, leaf blight, anthracnose, Powdery mildew, root rot, rust, yellow mosaic, etc.
- PC9.** identify the symptoms of pests and disease and determine the mode of pest and disease transfer
- PC10.** identify the part of the crop affected by pests and disease
- PC11.** determine the stage of pest and disease incidence along with the extent of damage and economic threshold levels (ETL) of the pests
- PC12.** check and document the pest
- PC13.** co-ordinate with the agricultural extension service agents and diagnostic clinics to determine the causal organism for the disease and its treatment

## Qualification Pack

### *Identify and apply the necessary treatment*

To be competent, the user/individual on the job must be able to:

- PC14.** identify natural enemies of the pulses pests such as ladybird, ground beetles, hoverfly, parasitoids and adopt them for pest control according to the advice of agricultural extension service workers, Krishi Vigyan Kendra (KVK), diagnostic clinics, etc.
- PC15.** select the appropriate Integrated Pest Management (IPM) techniques for removing the identified pests and disease such as trash mulching, de-trashing, egg destruction, bund freeing, water draining, usage of bio-control agents, etc.
- PC16.** apply the recommended treatment as per the prescription to remove pests and disease
- PC17.** follow the recommended safety practices while applying any treatment, such as using the relevant PPE
- PC18.** maintain the record of the use of any pesticides, insecticides and any other treatment

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** various types of diseases found in pulses and their symptoms
- KU2.** different biotic and abiotic factors causing diseases and disorders in plants
- KU3.** different mode of transmissions of disease such as implements, vectors, rain, wind
- KU4.** the importance of adopting safe production methods for safe produce
- KU5.** advantages of biological control of insects, pest & diseases, bio-pesticides and pheromones used in IPM (Integrated Pest Management)
- KU6.** minimum residue levels and Protected Health Information (PHI) for different types of pesticides
- KU7.** use of the pesticide spraying tools and equipment
- KU8.** national and international standards on pesticide residues
- KU9.** the benefits of using pest and disease-resistant varieties of pulses
- KU10.** recommended practices to restrict the entry of pathogens into the field through planting material, irrigation water, workers, tools and equipment, and vectors such as whitefly
- KU11.** the practice of crop rotation with suitable crops
- KU12.** the importance of identifying and removing the diseased crop to prevent the spread of pest and disease to the healthy crop
- KU13.** use of the recommended combination of biological, mechanical and chemical control methods for effective pest and disease prevention such as traps, sticky plates etc
- KU14.** how to identify different types of pests in pulses pod borers, hairy caterpillar, aphids, stem fly, whitefly, thrips, beetles, nematodes, etc.
- KU15.** how to identify plant disease vectors and major pulses diseases such as leaf spot, leaf blight, anthracnose, Powdery mildew, root rot, rust, yellow mosaic, etc.
- KU16.** the process of determining the stage of pest incidence along with the extent of damage and economic threshold levels (ETL) of the pests
- KU17.** use of IPM methods such as light and pheromone traps to identify the presence and population of insects and vectors
- KU18.** the process of determining the causal organism for the disease and its treatment

## Qualification Pack

- KU19.** natural enemies of the pest such as ladybird, ground beetles, hoverfly and the benefits of adopting them for pest control
- KU20.** the importance of applying the recommended treatment as per the prescription and maintaining the record of their use
- KU21.** the importance of using the recommended PPE while applying harmful chemicals
- KU22.** ways to minimise pollution caused due to overuse of pesticides
- KU23.** applicable record-keeping practices for effective pest and disease management such as the preparation of pest calendar
- KU24.** the process of preparing various bio-pesticides
- KU25.** the use of appropriate tools and equipment for pest and disease management

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write work-related notes
- GS2.** read the relevant literature to get the latest updates about the field of work
- GS3.** listen attentively to understand the information/ instructions being shared
- GS4.** communicate politely and professionally
- GS5.** plan and prioritise tasks to ensure timely completion
- GS6.** evaluate all possible solutions to a problem to select the best one
- GS7.** co-ordinate with the co-workers to achieve the work objectives
- GS8.** identify possible disruptions to work and take appropriate preventive measures
- GS9.** take quick decisions to deal with workplace emergencies/ accidents

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Follow the relevant preventive measures</i>                                                                                                                                                                                                                                          | <b>12</b>    | <b>16</b>       | -             | <b>8</b>   |
| <b>PC1.</b> use pest and disease-resistant varieties of pulses                                                                                                                                                                                                                          | -            | -               | -             | -          |
| <b>PC2.</b> carry out crop rotation with suitable crops                                                                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC3.</b> follow the recommended practices to restrict the entry of pathogens into the field through planting material, irrigation water, workers, tools and equipment, and vectors such as whitefly                                                                                  | -            | -               | -             | -          |
| <b>PC4.</b> follow the relevant preventive techniques according to the cultivar such as light trap, bird perch, sticky traps, etc.                                                                                                                                                      | -            | -               | -             | -          |
| <b>PC5.</b> identify and remove the diseased crop to prevent the spread of pest and disease to healthy crop                                                                                                                                                                             | -            | -               | -             | -          |
| <b>PC6.</b> use the recommended combination of biological, mechanical and chemical control methods for effective pest and disease prevention                                                                                                                                            | -            | -               | -             | -          |
| <i>Identify pests and disease in pulses</i>                                                                                                                                                                                                                                             | <b>8</b>     | <b>14</b>       | -             | <b>12</b>  |
| <b>PC7.</b> follow the recommended Integrated Pest Management (IPM) methods such as light and pheromone traps to identify the presence and population of pests, insects and vectors such as pod borers, hairy caterpillar, aphids, stem fly, whitefly, thrips, beetles, nematodes, etc. | -            | -               | -             | -          |
| <b>PC8.</b> conduct regular field visits to identify major pulse diseases such as leaf spot, leaf blight, anthracnose, Powdery mildew, root rot, rust, yellow mosaic, etc.                                                                                                              | -            | -               | -             | -          |
| <b>PC9.</b> identify the symptoms of pests and disease and determine the mode of pest and disease transfer                                                                                                                                                                              | -            | -               | -             | -          |
| <b>PC10.</b> identify the part of the crop affected by pests and disease                                                                                                                                                                                                                | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> determine the stage of pest and disease incidence along with the extent of damage and economic threshold levels (ETL) of the pests                                                                                                                           | -            | -               | -             | -          |
| <b>PC12.</b> check and document the pest                                                                                                                                                                                                                                  | -            | -               | -             | -          |
| <b>PC13.</b> co-ordinate with the agricultural extension service agents and diagnostic clinics to determine the causal organism for the disease and its treatment                                                                                                         | -            | -               | -             | -          |
| <i>Identify and apply the necessary treatment</i>                                                                                                                                                                                                                         | <b>10</b>    | <b>10</b>       | -             | <b>10</b>  |
| <b>PC14.</b> identify natural enemies of the pulses pests such as ladybird, ground beetles, hoverfly, parasitoids and adopt them for pest control according to the advice of agricultural extension service workers, Krishi Vigyan Kendra (KVK), diagnostic clinics, etc. | -            | -               | -             | -          |
| <b>PC15.</b> select the appropriate Integrated Pest Management (IPM) techniques for removing the identified pests and disease such as trash mulching, de-trashing, egg destruction, bund freeing, water draining, usage of bio-control agents, etc.                       | -            | -               | -             | -          |
| <b>PC16.</b> apply the recommended treatment as per the prescription to remove pests and disease                                                                                                                                                                          | -            | -               | -             | -          |
| <b>PC17.</b> follow the recommended safety practices while applying any treatment, such as using the relevant PPE                                                                                                                                                         | -            | -               | -             | -          |
| <b>PC18.</b> maintain the record of the use of any pesticides, insecticides and any other treatment                                                                                                                                                                       | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                                                                          | <b>30</b>    | <b>40</b>       | -             | <b>30</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                           |
|----------------------------|-----------------------------------------------------------|
| <b>NOS Code</b>            | AGR/N0110                                                 |
| <b>NOS Name</b>            | Perform integrated pest and disease management for pulses |
| <b>Sector</b>              | Agriculture                                               |
| <b>Sub-Sector</b>          | Agriculture Crop Production                               |
| <b>Occupation</b>          | Field Crops Cultivation (Food Crops)                      |
| <b>NSQF Level</b>          | 3                                                         |
| <b>Credits</b>             | 1                                                         |
| <b>Version</b>             | 3.0                                                       |
| <b>Last Reviewed Date</b>  | 18/02/2025                                                |
| <b>Next Review Date</b>    | 18/02/2028                                                |
| <b>NSQC Clearance Date</b> | 18/02/2025                                                |

## Qualification Pack

### AGR/N0111: Perform irrigation management for field crops

#### Description

This OS unit is about performing irrigation management of field crops.

#### Scope

The scope covers the following :

- Prepare for field irrigation
- Irrigate the field
- Manage the water usage

#### Elements and Performance Criteria

##### *Prepare for field irrigation*

To be competent, the user/individual on the job must be able to:

- PC1.** collect a water sample from the source of irrigation and coordinate with an authorised lab to get it tested
- PC2.** follow the measures recommended by the lab to improve the water quality
- PC3.** set up an appropriate irrigation system such as surface irrigation, drip irrigation, sub-surface irrigation system based on the requirement of the specific field crop

##### *Irrigate the field*

To be competent, the user/individual on the job must be able to:

- PC4.** irrigate the field according to the recommended irrigation schedule for the crop, ensuring there is adequate water supply at various stages of crop's growth
- PC5.** maintain the record of field irrigation to ensure irrigation as per the schedule
- PC6.** follow the recommended practices to prevent over and under-irrigation, ensuring there is no waterlogging at any stage of the crop's growth

##### *Manage the water usage*

To be competent, the user/individual on the job must be able to:

- PC7.** follow the recommended practices for effective drainage of excess water from the field
- PC8.** maintain the recommended level of water in the soil to prevent the harmful effects of inappropriate levels of moisture in it
- PC9.** plug water spills and leakages to prevent its wastage
- PC10.** use efficient irrigation techniques to minimize water waste and conserve resources
- PC11.** integrate water-saving irrigation practices like drip irrigation and rainwater harvesting

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the timing and method of irrigation appropriate for a given soil type and climatic conditions

## Qualification Pack

- KU2.** the quantity of water required for different types of crops and its effect on the yield
- KU3.** the importance of sampling and testing irrigation water through an authorised lab to determine its quality
- KU4.** various measures to be followed to improve the water quality
- KU5.** the process of setting up different types of irrigation systems such as surface irrigation, drip irrigation, sub-surface irrigation system, etc.
- KU6.** the advantages and disadvantages of using different types of irrigation systems
- KU7.** the importance of irrigating the field according to the recommended irrigation schedule for the crop and the factors to be considered in scheduling irrigation
- KU8.** the recommended practices to be followed to prevent over and under-irrigation
- KU9.** the recommended practices to be followed for effective drainage of excess water from the field
- KU10.** the importance of maintaining the recommended level of water in the soil to prevent the harmful effects caused by inappropriate levels of moisture
- KU11.** various practices to be followed to optimise the usage of water and prevent its wastage

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write work-related notes
- GS2.** read the relevant guides, manuals and literature to get the latest information about the field of work
- GS3.** communicate politely and professionally
- GS4.** listen attentively to understand the instructions being given
- GS5.** identify solutions to work-related issues
- GS6.** plan and prioritise tasks to ensure timely completion
- GS7.** take quick decisions to deal with any emergencies or accidents
- GS8.** plan effective use of time and resources

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Prepare for field irrigation</i>                                                                                                                                                | <b>12</b>    | <b>16</b>       | -             | <b>10</b>  |
| <b>PC1.</b> collect a water sample from the source of irrigation and coordinate with an authorised lab to get it tested                                                            | -            | -               | -             | -          |
| <b>PC2.</b> follow the measures recommended by the lab to improve the water quality                                                                                                | -            | -               | -             | -          |
| <b>PC3.</b> set up an appropriate irrigation system such as surface irrigation, drip irrigation, sub-surface irrigation system based on the requirement of the specific field crop | -            | -               | -             | -          |
| <i>Irrigate the field</i>                                                                                                                                                          | <b>8</b>     | <b>12</b>       | -             | <b>12</b>  |
| <b>PC4.</b> irrigate the field according to the recommended irrigation schedule for the crop, ensuring there is adequate water supply at various stages of crop's growth           | -            | -               | -             | -          |
| <b>PC5.</b> maintain the record of field irrigation to ensure irrigation as per the schedule                                                                                       | -            | -               | -             | -          |
| <b>PC6.</b> follow the recommended practices to prevent over and under-irrigation, ensuring there is no waterlogging at any stage of the crop's growth                             | -            | -               | -             | -          |
| <i>Manage the water usage</i>                                                                                                                                                      | <b>10</b>    | <b>12</b>       | -             | <b>8</b>   |
| <b>PC7.</b> follow the recommended practices for effective drainage of excess water from the field                                                                                 | -            | -               | -             | -          |
| <b>PC8.</b> maintain the recommended level of water in the soil to prevent the harmful effects of inappropriate levels of moisture in it                                           | -            | -               | -             | -          |
| <b>PC9.</b> plug water spills and leakages to prevent its wastage                                                                                                                  | -            | -               | -             | -          |
| <b>PC10.</b> use efficient irrigation techniques to minimize water waste and conserve resources                                                                                    | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> integrate water-saving irrigation practices like drip irrigation and rainwater harvesting | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                       | <b>30</b>    | <b>40</b>       | <b>-</b>      | <b>30</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                               |
|----------------------------|-----------------------------------------------|
| <b>NOS Code</b>            | AGR/N0111                                     |
| <b>NOS Name</b>            | Perform irrigation management for field crops |
| <b>Sector</b>              | Agriculture                                   |
| <b>Sub-Sector</b>          | Agriculture Crop Production                   |
| <b>Occupation</b>          | Field Crops Cultivation (Food Crops)          |
| <b>NSQF Level</b>          | 4                                             |
| <b>Credits</b>             | 1                                             |
| <b>Version</b>             | 4.0                                           |
| <b>Last Reviewed Date</b>  | 08/05/2025                                    |
| <b>Next Review Date</b>    | 31/10/2025                                    |
| <b>NSQC Clearance Date</b> | 08/05/2025                                    |

## Qualification Pack

# AGR/N0120: Carry out harvesting, post-harvest processing and marketing of pulses

## Description

This OS unit is about harvesting, post-harvest processing and marketing pulses.

## Scope

The scope covers the following :

- Harvest the pulses
- Thresh and pack the pulses
- Market the pulses

## Elements and Performance Criteria

### Harvest the pulses

To be competent, the user/individual on the job must be able to:

- PC1.** check the maturity indicators of the pulse crop to ensure its readiness for being harvested
- PC2.** ascertain the moisture content in the crop to see if it is appropriate for harvesting
- PC3.** select the manual or mechanised harvesting method according to the quantity of crop to be harvested
- PC4.** arrange the necessary tools, equipment and machinery for harvesting pulses and prepare them for use
- PC5.** treat the crop with the appropriate chemicals such as defoliant before harvesting, as applicable
- PC6.** harvest the pulses crop at the appropriate stage and time, avoiding pod opening and ensuring minimum loss
- PC7.** dry harvested pulses under sunlight to reduce moisture content and prevent spoilage

### Thresh and pack the pulses

To be competent, the user/individual on the job must be able to:

- PC8.** thresh the harvested pulses to obtain the seeds and ensure pulses don't catch moisture
- PC9.** carry out cleaning of the harvested pulses using appropriate methods and dry them under the sun or mechanically
- PC10.** sort and grade the cleaned pulses on the basis of applicable parameters
- PC11.** select a dry storage area with the recommended temperature and humidity for storing pulses
- PC12.** prepare the storage area, ensuring it is hygienic, pest and rodent-free
- PC13.** select appropriate packing material for packing pulses such as jute bags, Polypropylene (PP) pouches High-Density Polyethylene (HDPE) packaging, etc.
- PC14.** pack pulses following the recommended measures to prevent the absorption of moisture

### Market the pulses

To be competent, the user/individual on the job must be able to:

## Qualification Pack

- PC15.** select an appropriate time for selling pulses based on the periodical demand of the produce and profitability
- PC16.** identify the buyers of pulses such as eMandi, procurement agencies, traders, co-operatives, etc.
- PC17.** coordinate and negotiate with the buyer to secure a profitable price for the produce
- PC18.** arrange an appropriate mode of transport to deliver pulses to the buyer
- PC19.** process the payments using the buyer-preferred e-payment method
- PC20.** calculate the benefit-cost (B:C) ratio
- PC21.** maintain the manual and/ or electronic record of sales and payments using the physical registers and/ or the relevant computer application

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the ideal climate and conditions for harvesting pulses
- KU2.** how to assess the maturity of pulses to ensure their readiness for being harvested
- KU3.** the manual and mechanical methods of harvesting pulses and the use of relevant tools, implements and equipment
- KU4.** the importance and ways of protecting the harvested pulses from catching moisture
- KU5.** different crop cleaning and threshing techniques
- KU6.** the process of threshing harvested pulses to obtain the seeds
- KU7.** the appropriate temperature and humidity required for storing the harvested pulses
- KU8.** the criteria for sorting and grading pulses
- KU9.** the relevant methods to avoid crop loss due to moisture
- KU10.** various methods of storage, their cost dynamics and influence on the quality of pulses
- KU11.** the chemical-based and other relevant methods to prevent losses from pests and rodents in the storage area
- KU12.** various storage pests, their behaviour and management
- KU13.** the appropriate packing material to pack pulses such as jute bags, Polypropylene (PP) pouches High-Density Polyethylene (HDPE) packaging, etc.
- KU14.** the process of identifying and negotiating with potential buyers
- KU15.** the appropriate mode of transport for transporting pulses
- KU16.** how to use various e-payment methods
- KU17.** how to calculate the benefit-cost (B:C) ratio
- KU18.** how to maintain manual and electronic records

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** read the relevant literature to get the latest updates about the field of work

## Qualification Pack

- GS3.** communicate clearly and politely
- GS4.** listen attentively to understand the instructions being given
- GS5.** identify appropriate solutions to work-related problems
- GS6.** plan and prioritise tasks to ensure timely completion
- GS7.** take quick decisions to deal with any emergencies/ accidents
- GS8.** plan effective use of time and resources

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                            | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Harvest the pulses</i>                                                                                                   | <b>8</b>     | <b>16</b>       | -             | <b>8</b>   |
| <b>PC1.</b> check the maturity indicators of the pulse crop to ensure its readiness for being harvested                     | -            | -               | -             | -          |
| <b>PC2.</b> ascertain the moisture content in the crop to see if it is appropriate for harvesting                           | -            | -               | -             | -          |
| <b>PC3.</b> select the manual or mechanised harvesting method according to the quantity of crop to be harvested             | -            | -               | -             | -          |
| <b>PC4.</b> arrange the necessary tools, equipment and machinery for harvesting pulses and prepare them for use             | -            | -               | -             | -          |
| <b>PC5.</b> treat the crop with the appropriate chemicals such as defoliants before harvesting, as applicable               | -            | -               | -             | -          |
| <b>PC6.</b> harvest the pulses crop at the appropriate stage and time, avoiding pod opening and ensuring minimum loss       | -            | -               | -             | -          |
| <b>PC7.</b> dry harvested pulses under sunlight to reduce moisture content and prevent spoilage                             | -            | -               | -             | -          |
| <i>Thresh and pack the pulses</i>                                                                                           | <b>8</b>     | <b>10</b>       | -             | <b>10</b>  |
| <b>PC8.</b> thresh the harvested pulses to obtain the seeds and ensure pulses don't catch moisture                          | -            | -               | -             | -          |
| <b>PC9.</b> carry out cleaning of the harvested pulses using appropriate methods and dry them under the sun or mechanically | -            | -               | -             | -          |
| <b>PC10.</b> sort and grade the cleaned pulses on the basis of applicable parameters                                        | -            | -               | -             | -          |
| <b>PC11.</b> select a dry storage area with the recommended temperature and humidity for storing pulses                     | -            | -               | -             | -          |
| <b>PC12.</b> prepare the storage area, ensuring it is hygienic, pest and rodent-free                                        | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> select appropriate packing material for packing pulses such as jute bags, Polypropylene (PP) pouches High-Density Polyethylene (HDPE) packaging, etc. | -            | -               | -             | -          |
| <b>PC14.</b> pack pulses following the recommended measures to prevent the absorption of moisture                                                                  | -            | -               | -             | -          |
| <i>Market the pulses</i>                                                                                                                                           | <b>14</b>    | <b>14</b>       | -             | <b>12</b>  |
| <b>PC15.</b> select an appropriate time for selling pulses based on the periodical demand of the produce and profitability                                         | -            | -               | -             | -          |
| <b>PC16.</b> identify the buyers of pulses such as eMandi, procurement agencies, traders, co-operatives, etc.                                                      | -            | -               | -             | -          |
| <b>PC17.</b> coordinate and negotiate with the buyer to secure a profitable price for the produce                                                                  | -            | -               | -             | -          |
| <b>PC18.</b> arrange an appropriate mode of transport to deliver pulses to the buyer                                                                               | -            | -               | -             | -          |
| <b>PC19.</b> process the payments using the buyer-preferred e-payment method                                                                                       | -            | -               | -             | -          |
| <b>PC20.</b> calculate the benefit-cost (B:C) ratio                                                                                                                | -            | -               | -             | -          |
| <b>PC21.</b> maintain the manual and/ or electronic record of sales and payments using the physical registers and/ or the relevant computer application            | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                   | <b>30</b>    | <b>40</b>       | -             | <b>30</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                       |
|----------------------------|-----------------------------------------------------------------------|
| <b>NOS Code</b>            | AGR/N0120                                                             |
| <b>NOS Name</b>            | Carry out harvesting, post-harvest processing and marketing of pulses |
| <b>Sector</b>              | Agriculture                                                           |
| <b>Sub-Sector</b>          | Agriculture Crop Production                                           |
| <b>Occupation</b>          | Field Crops Cultivation (Food Crops)                                  |
| <b>NSQF Level</b>          | 3                                                                     |
| <b>Credits</b>             | 1                                                                     |
| <b>Version</b>             | 3.0                                                                   |
| <b>Last Reviewed Date</b>  | 18/02/2025                                                            |
| <b>Next Review Date</b>    | 18/02/2028                                                            |
| <b>NSQC Clearance Date</b> | 18/02/2025                                                            |

## Qualification Pack

### AGR/N9933: Participate in collective farming activities

#### Description

This OS unit is about participating in collective farming activities in Producer Groups (PGs), Farmers Interest Groups (FIGs), Self-Help Groups (SHGs) and other similar groups to attain a common objective.

#### Scope

The scope covers the following :

- Create PGs/ FIGs/ SHGs
- Prepare for the PG/ FIG/ SHG operations
- Conduct group meetings and training sessions
- Carry out collective farming/ activities

#### Elements and Performance Criteria

##### *Create PGs/ FIGs/ SHGs*

To be competent, the user/individual on the job must be able to:

- PC1.** identify farmers/ groups with the common interests in the area
- PC2.** create Producer Groups (PGs)/Farmers Interest Groups (FIGs)/ Self-Help Groups (SHGs), following the applicable rules and regulations

##### *Prepare for the PG/ FIG/ SHG operations*

To be competent, the user/individual on the job must be able to:

- PC3.** organise fundraising activities to support the functioning of the group
- PC4.** establish links with the local government at panchayat level to obtain access to the relevant development programmes and funds
- PC5.** induct subject matter experts (SMEs) in the group
- PC6.** assist in arranging the required Information and Communication Technology (ICT) products for the group
- PC7.** plan the commodity convergence with the relevant developmental programmes
- PC8.** plan optimal production to meet the market and household food security needs

##### *Conduct group meetings and training sessions*

To be competent, the user/individual on the job must be able to:

- PC9.** conduct the initial group meetings to introduce the members, discuss the group objectives, group income-generating enterprises/ activities, methods of operation, etc.
- PC10.** assist in exchanging the domain and technical knowledge such as market or price information, latest technology, and resolving common issues or conflicts through the PG/ FIG/ SHG meetings
- PC11.** organise capacity building exercises such as skill development and training programmes

##### *Carry out collective farming/ activities*

To be competent, the user/individual on the job must be able to:

## Qualification Pack

- PC12.** organise field trials to identify and resolve problems encountered by group members in the field operations
- PC13.** procure/hire advanced and expensive farm machineries/tools and equipment using the group fund for collective use of the group members
- PC14.** establish and manage the group-owned bank of quality seeds/ fertilisers/ pesticides/ tools and equipment, etc.
- PC15.** use the group's credit facility as per the applicable terms and conditions
- PC16.** carry out relevant duties as per own role in the PG/FIG/ SHG such as the group leader/ secretary/ book-keeper, etc.
- PC17.** co-ordinate within the group(s) in procuring inputs in bulk/large-scale farming, packing/transportation/marketing of the produce, etc.
- PC18.** assist in forming forward and backward linkages through the PGs/ FIGs/ SHGs
- PC19.** identify and follow the relevant practices to add value to the produce such as processing, packing, upgrading the quality, etc.
- PC20.** arrange for the regular repair and maintenance of the farm machineries/tools, equipment/tube/bore wells/storage/drying platforms/processing units, etc.
- PC21.** connect and partner with other groups to expand the network and address common problems at a large scale

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the process of preparing for the PG/ FIG/ SHG operations such as fundraising, induction of SMEs, investing in ICT products, etc.
- KU2.** how to establish links with the local government at the panchayat level to obtain access to the relevant development programmes and funds
- KU3.** the process of commodity convergence with the relevant developmental programmes
- KU4.** the importance of planning optimal production to meet the market and household food security needs
- KU5.** the importance of setting the group objectives and deciding the group income-generating enterprises/ activities, methods of operation, benefits, etc.
- KU6.** the importance of organising the PG/FIG/ SHG meetings and training sessions to resolve common concerns and get information about the latest developments in the field of work
- KU7.** the benefits of various capacity building exercises such as skill development and training programmes
- KU8.** the importance and process of conducting field trials to identify and resolve problems encountered by farmers in the field operations
- KU9.** the concept of group-owned bank to provide quality seeds, fertilisers, pesticides, tools and equipment to the member farmers
- KU10.** the process of using the group's credit facility
- KU11.** core collective farming activities such as procuring inputs in bulk, large-scale farming, packing, transportation and marketing of the produce, etc.
- KU12.** the concept and benefits of forming forward and backward linkages

## Qualification Pack

- KU13.** relevant value addition practices such as processing, packing, upgrading the quality, etc.
- KU14.** the benefits of connecting with similar groups to address common problems at a large scale

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write relevant notes and reports
- GS2.** read the relevant literature to get latest updates about the field of work
- GS3.** communicate politely and professionally
- GS4.** listen attentively to understand the information being shared
- GS5.** plan tasks for effective use of time
- GS6.** identify possible disruptions to work and take appropriate preventive measures
- GS7.** evaluate all possible solutions to a problem to select the best one

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Create PGs/ FIGs/ SHGs</i>                                                                                                                                                                           | <b>6</b>     | <b>8</b>        | -             | <b>6</b>   |
| <b>PC1.</b> identify farmers/ groups with the common interests in the area                                                                                                                              | -            | -               | -             | -          |
| <b>PC2.</b> create Producer Groups (PGs)/Farmers Interest Groups (FIGs)/ Self-Help Groups (SHGs), following the applicable rules and regulations                                                        | -            | -               | -             | -          |
| <i>Prepare for the PG/ FIG/ SHG operations</i>                                                                                                                                                          | <b>6</b>     | <b>10</b>       | -             | <b>6</b>   |
| <b>PC3.</b> organise fundraising activities to support the functioning of the group                                                                                                                     | -            | -               | -             | -          |
| <b>PC4.</b> establish links with the local government at panchayat level to obtain access to the relevant development programmes and funds                                                              | -            | -               | -             | -          |
| <b>PC5.</b> induct subject matter experts (SMEs) in the group                                                                                                                                           | -            | -               | -             | -          |
| <b>PC6.</b> assist in arranging the required Information and Communication Technology (ICT) products for the group                                                                                      | -            | -               | -             | -          |
| <b>PC7.</b> plan the commodity convergence with the relevant developmental programmes                                                                                                                   | -            | -               | -             | -          |
| <b>PC8.</b> plan optimal production to meet the market and household food security needs                                                                                                                | -            | -               | -             | -          |
| <i>Conduct group meetings and training sessions</i>                                                                                                                                                     | <b>8</b>     | <b>6</b>        | -             | <b>8</b>   |
| <b>PC9.</b> conduct the initial group meetings to introduce the members, discuss the group objectives, group income-generating enterprises/ activities, methods of operation, etc.                      | -            | -               | -             | -          |
| <b>PC10.</b> assist in exchanging the domain and technical knowledge such as market or price information, latest technology, and resolving common issues or conflicts through the PG/ FIG/ SHG meetings | -            | -               | -             | -          |
| <b>PC11.</b> organise capacity building exercises such as skill development and training programmes                                                                                                     | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Carry out collective farming/ activities</i>                                                                                                                      | <b>10</b>    | <b>16</b>       | <b>-</b>      | <b>10</b>  |
| <b>PC12.</b> organise field trials to identify and resolve problems encountered by group members in the field operations                                             | -            | -               | -             | -          |
| <b>PC13.</b> procure/hire advanced and expensive farm machineries/tools and equipment using the group fund for collective use of the group members                   | -            | -               | -             | -          |
| <b>PC14.</b> establish and manage the group-owned bank of quality seeds/ fertilisers/ pesticides/ tools and equipment, etc.                                          | -            | -               | -             | -          |
| <b>PC15.</b> use the group's credit facility as per the applicable terms and conditions                                                                              | -            | -               | -             | -          |
| <b>PC16.</b> carry out relevant duties as per own role in the PG/FIG/ SHG such as the group leader/ secretary/ book-keeper, etc.                                     | -            | -               | -             | -          |
| <b>PC17.</b> co-ordinate within the group(s) in procuring inputs in bulk/large-scale farming, packing/transportation/marketing of the produce, etc.                  | -            | -               | -             | -          |
| <b>PC18.</b> assist in forming forward and backward linkages through the PGs/ FIGs/ SHGs                                                                             | -            | -               | -             | -          |
| <b>PC19.</b> identify and follow the relevant practices to add value to the produce such as processing, packing, upgrading the quality, etc.                         | -            | -               | -             | -          |
| <b>PC20.</b> arrange for the regular repair and maintenance of the farm machineries/tools, equipment/tube/bore wells/storage/drying platforms/processing units, etc. | -            | -               | -             | -          |
| <b>PC21.</b> connect and partner with other groups to expand the network and address common problems at a large scale                                                | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                     | <b>30</b>    | <b>40</b>       | <b>-</b>      | <b>30</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                              |
|----------------------------|----------------------------------------------|
| <b>NOS Code</b>            | AGR/N9933                                    |
| <b>NOS Name</b>            | Participate in collective farming activities |
| <b>Sector</b>              | Agriculture                                  |
| <b>Sub-Sector</b>          | Generic                                      |
| <b>Occupation</b>          | Generic                                      |
| <b>NSQF Level</b>          | 3                                            |
| <b>Credits</b>             | 1                                            |
| <b>Version</b>             | 1.0                                          |
| <b>Last Reviewed Date</b>  | 18/02/2025                                   |
| <b>Next Review Date</b>    | 18/02/2028                                   |
| <b>NSQC Clearance Date</b> | 18/02/2025                                   |

## Qualification Pack

### AGR/N9932: Follow health and safety guidelines

#### Description

This OS is about maintaining health and safety of self and other co-workers at the worksite

#### Scope

The scope covers the following :

- Maintain personal hygiene
- Maintain clean and safe workplace
- Administer appropriate emergency procedures

#### Elements and Performance Criteria

##### *Maintain personal hygiene*

To be competent, the user/individual on the job must be able to:

- PC1.** wash hands, legs and face with soap/alcohol based sanitizer at reasonable intervals
- PC2.** wash the worn clothes with soap and sun dry before use next time
- PC3.** ensure the face is covered with mask or three layers of cloth-piece

##### *Maintain clean and safe workplace*

To be competent, the user/individual on the job must be able to:

- PC4.** carry out basic safety checks before operation of all tools, implements, and machinery and report identified hazards to the supervisor
- PC5.** wear appropriate Personal Protective Equipment (PPE) while performing work in accordance with the workplace policy
- PC6.** follow the instructions mentioned on the labels of chemicals/pesticides/fumigants etc to avoid hazards
- PC7.** assess risks prior to performing manual handling jobs, and work according to currently recommended safe practices
- PC8.** sanitize equipment, tools and machinery before and after use
- PC9.** use equipment and materials safely and correctly and return the same to designated storage after use
- PC10.** dispose waste safely and correctly in the designated area
- PC11.** work in a manner which minimizes environmental damage, ensuring all procedures and instructions for controlling risks are followed
- PC12.** report any accidents, incidents or problems without delay to an appropriate person and take necessary immediate action to reduce further danger
- PC13.** follow government / workplace advisories incase of outbreak of any disease/disaster

##### *Administer appropriate emergency procedures*

To be competent, the user/individual on the job must be able to:

## Qualification Pack

- PC14.** follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to the location of emergency, as per the workplace requirements
- PC15.** provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques
- PC16.** recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate
- PC17.** report details of first aid administered in accordance with workplace procedures

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant legislation, standards, policies, and procedures at work
- KU2.** relevant health and safety requirements applicable to the work environment
- KU3.** own job role and responsibilities and sources of information pertaining to work
- KU4.** who to approach for support in order to obtain work related information, clarifications and support
- KU5.** importance of following health, hygiene, safety and quality standards and the impact of not following the standards on consumers and the business
- KU6.** personal hygiene and fitness requirement
- KU7.** importance of sanitization of the workplace
- KU8.** types of Personal Protective Equipment (PPE) required at the workplace and their importance
- KU9.** the correct and safe way to use materials and equipment required for the work
- KU10.** safe waste disposal methods
- KU11.** methods for minimizing environmental damage during work
- KU12.** the risks to health and safety including contagious diseases and the measures to be taken to control those risks in the area of work
- KU13.** workplace procedures and requirements for the prevention and treatment of workplace injuries/illnesses.
- KU14.** basic emergency first aid procedure
- KU15.** local emergency services
- KU16.** why accidents, incidents and problems should be reported and the appropriate actions to be taken

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record the data as per the requirement
- GS2.** report problems to the appropriate personnel in a timely manner
- GS3.** read instruction manual for hand tool and equipments
- GS4.** communicate clearly and effectively with co-workers, and other stakeholders

## Qualification Pack

- GS5.** comprehend information shared by senior people and experts
- GS6.** make decisions pertaining to personal hygiene and safety
- GS7.** schedule daily activities and draw up priorities
- GS8.** manage relationships with co-workers, manager and other stakeholders
- GS9.** assess situation and identify appropriate control measures

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Maintain personal hygiene</i>                                                                                                                   | <b>10</b>    | <b>5</b>        | -             | <b>10</b>  |
| <b>PC1.</b> wash hands, legs and face with soap/alcohol based sanitizer at reasonable intervals                                                    | -            | -               | -             | -          |
| <b>PC2.</b> wash the worn clothes with soap and sun dry before use next time                                                                       | -            | -               | -             | -          |
| <b>PC3.</b> ensure the face is covered with mask or three layers of cloth-piece                                                                    | -            | -               | -             | -          |
| <i>Maintain clean and safe workplace</i>                                                                                                           | <b>15</b>    | <b>15</b>       | -             | <b>15</b>  |
| <b>PC4.</b> carry out basic safety checks before operation of all tools, implements, and machinery and report identified hazards to the supervisor | -            | -               | -             | -          |
| <b>PC5.</b> wear appropriate Personal Protective Equipment (PPE) while performing work in accordance with the workplace policy                     | -            | -               | -             | -          |
| <b>PC6.</b> follow the instructions mentioned on the labels of chemicals/pesticides/fumigants etc to avoid hazards                                 | -            | -               | -             | -          |
| <b>PC7.</b> assess risks prior to performing manual handling jobs, and work according to currently recommended safe practices                      | -            | -               | -             | -          |
| <b>PC8.</b> sanitize equipment, tools and machinery before and after use                                                                           | -            | -               | -             | -          |
| <b>PC9.</b> use equipment and materials safely and correctly and return the same to designated storage after use                                   | -            | -               | -             | -          |
| <b>PC10.</b> dispose waste safely and correctly in the designated area                                                                             | -            | -               | -             | -          |
| <b>PC11.</b> work in a manner which minimizes environmental damage, ensuring all procedures and instructions for controlling risks are followed    | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> report any accidents, incidents or problems without delay to an appropriate person and take necessary immediate action to reduce further danger                                      | -            | -               | -             | -          |
| <b>PC13.</b> follow government / workplace advisories incase of outbreak of any disease/disaster                                                                                                  | -            | -               | -             | -          |
| <i>Administer appropriate emergency procedures</i>                                                                                                                                                | <b>15</b>    | <b>5</b>        | -             | <b>10</b>  |
| <b>PC14.</b> follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to the location of emergency, as per the workplace requirements | -            | -               | -             | -          |
| <b>PC15.</b> provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques                                                                           | -            | -               | -             | -          |
| <b>PC16.</b> recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate                                                                     | -            | -               | -             | -          |
| <b>PC17.</b> report details of first aid administered in accordance with workplace procedures                                                                                                     | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                  | <b>40</b>    | <b>25</b>       | -             | <b>35</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                     |
|----------------------------|-------------------------------------|
| <b>NOS Code</b>            | AGR/N9932                           |
| <b>NOS Name</b>            | Follow health and safety guidelines |
| <b>Sector</b>              | Agriculture                         |
| <b>Sub-Sector</b>          | Generic                             |
| <b>Occupation</b>          | Generic                             |
| <b>NSQF Level</b>          | 3                                   |
| <b>Credits</b>             | 1                                   |
| <b>Version</b>             | 1.0                                 |
| <b>Last Reviewed Date</b>  | 18/02/2025                          |
| <b>Next Review Date</b>    | 18/02/2028                          |
| <b>NSQC Clearance Date</b> | 18/02/2025                          |

## Qualification Pack

### DGT/VSQ/N0101: Employability Skills (30 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

**PC1.** understand the significance of employability skills in meeting the job requirements

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

**PC2.** identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

**PC3.** explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

**PC4.** speak with others using some basic English phrases or sentences

##### *Communication Skills*

To be competent, the user/individual on the job must be able to:

**PC5.** follow good manners while communicating with others

**PC6.** work with others in a team

## Qualification Pack

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

**PC7.** communicate and behave appropriately with all genders and PwD

**PC8.** report any issues related to sexual harassment

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

**PC9.** use various financial products and services safely and securely

**PC10.** calculate income, expenses, savings etc.

**PC11.** approach the concerned authorities for any exploitation as per legal rights and laws

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

**PC12.** operate digital devices and use its features and applications securely and safely

**PC13.** use internet and social media platforms securely and safely

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

**PC14.** identify and assess opportunities for potential business

**PC15.** identify sources for arranging money and associated financial and legal challenges

### *Customer Service*

To be competent, the user/individual on the job must be able to:

**PC16.** identify different types of customers

**PC17.** identify customer needs and address them appropriately

**PC18.** follow appropriate hygiene and grooming standards

### *Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC19.** create a basic biodata

**PC20.** search for suitable jobs and apply

**PC21.** identify and register apprenticeship opportunities as per requirement

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use basic spoken English language

**KU6.** Do and dont of effective communication

**KU7.** inclusivity and its importance

**KU8.** different types of disabilities and appropriate communication and behaviour towards PwD

**KU9.** different types of financial products and services

## Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> understand the significance of employability skills in meeting the job requirements                                                                                                                                                                | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                     | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC2.</b> identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                             | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC3.</b> explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc. | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                    | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC4.</b> speak with others using some basic English phrases or sentences                                                                                                                                                                                    | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC5.</b> follow good manners while communicating with others                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC6.</b> work with others in a team                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                                                                                                                                                               | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC7.</b> communicate and behave appropriately with all genders and PwD                                                                                                                                                                                      | -            | -               | -             | -          |
| <b>PC8.</b> report any issues related to sexual harassment                                                                                                                                                                                                     | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                                                                                                                                                            | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC9.</b> use various financial products and services safely and securely                                                                                                                                                                                    | -            | -               | -             | -          |

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| Assessment Criteria for Outcomes                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> calculate income, expenses, savings etc.                                             | -            | -               | -             | -          |
| <b>PC11.</b> approach the concerned authorities for any exploitation as per legal rights and laws | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                   | <b>4</b>     | <b>6</b>        | -             | -          |
| <b>PC12.</b> operate digital devices and use its features and applications securely and safely    | -            | -               | -             | -          |
| <b>PC13.</b> use internet and social media platforms securely and safely                          | -            | -               | -             | -          |
| <i>Entrepreneurship</i>                                                                           | <b>3</b>     | <b>5</b>        | -             | -          |
| <b>PC14.</b> identify and assess opportunities for potential business                             | -            | -               | -             | -          |
| <b>PC15.</b> identify sources for arranging money and associated financial and legal challenges   | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC16.</b> identify different types of customers                                                | -            | -               | -             | -          |
| <b>PC17.</b> identify customer needs and address them appropriately                               | -            | -               | -             | -          |
| <b>PC18.</b> follow appropriate hygiene and grooming standards                                    | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC19.</b> create a basic biodata                                                               | -            | -               | -             | -          |
| <b>PC20.</b> search for suitable jobs and apply                                                   | -            | -               | -             | -          |
| <b>PC21.</b> identify and register apprenticeship opportunities as per requirement                | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                  | <b>20</b>    | <b>30</b>       | -             | -          |

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### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0101                   |
| <b>NOS Name</b>            | Employability Skills (30 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 2                               |
| <b>Credits</b>             | 1                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 06/02/2026                      |
| <b>Next Review Date</b>    | 06/02/2029                      |
| <b>NSQC Clearance Date</b> | 06/02/2026                      |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on a knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
5. In case of successfully passing only a certain number of NOSs, the trainee is eligible to take a

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subsequent assessment on the balance NOS's to pass the Qualification Pack.

6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

### Minimum Aggregate Passing % at QP Level : 50

**(Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

## Assessment Weightage

### Compulsory NOS

| National Occupational Standards                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage |
|---------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|-----------|
| AGR/N0118.Select and prepare the site and sow the pulse seeds                   | 30           | 40              | -             | 30         | 100         | 15        |
| AGR/N0108.Carry out macro and micronutrient management for field crops          | 30           | 40              | -             | 30         | 100         | 15        |
| AGR/N0109.Manage weed growth in crop fields                                     | 30           | 40              | -             | 30         | 100         | 10        |
| AGR/N0110.Perform integrated pest and disease management for pulses             | 30           | 40              | -             | 30         | 100         | 15        |
| AGR/N0111.Perform irrigation management for field crops                         | 30           | 40              | -             | 30         | 100         | 10        |
| AGR/N0120.Carry out harvesting, post-harvest processing and marketing of pulses | 30           | 40              | -             | 30         | 100         | 20        |
| AGR/N9933.Participate in collective farming activities                          | 30           | 40              | -             | 30         | 100         | 5         |
| AGR/N9932.Follow health and safety guidelines                                   | 40           | 25              | -             | 35         | 100         | 5         |

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| National Occupational Standards               | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|-----------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| DGT/VSQ/N0101.Employability Skills (30 Hours) | 20           | 30              | -             | -          | 50          | 5          |
| <b>Total</b>                                  | <b>270</b>   | <b>335</b>      | <b>-</b>      | <b>245</b> | <b>850</b>  | <b>100</b> |

## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

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|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |